# DS Project

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```
library(haven)
brfss <- read_xpt("/Users/aadityasingh/Downloads/Data Science Project/LLCP2024.XPT ")
# Note the trailing space at the end of the filename
```

```
dim(brfss)
```

```
## [1] 457670    301
```

**Filtering states where social determinant questions were not asked: SDLONELY, EMTSPRT,LSATISFY**

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
brfss %>%
  filter(!is.na(EMTSUPRT)) %>%
  distinct(`_STATE`) %>%
  arrange(`_STATE`)
```

```
## # A tibble: 33 x 1
##    `_STATE`
##       <dbl>
##  1        1
##  2        2
##  3        4
##  4        5
##  5        6
##  6       10
##  7       11
##  8       13
##  9       15
## 10       16
## # i 23 more rows
```

```
library(dplyr)
```

```r
brfss %>%
  filter(!is.na(LSATISFY)) %>%
  distinct(`_STATE`) %>%
  arrange(`_STATE`)
```

```
## # A tibble: 33 x 1
##    `_STATE`
##       <dbl>
##  1        1
##  2        2
##  3        4
##  4        5
##  5        6
##  6       10
##  7       11
##  8       13
##  9       15
## 10       16
## # i 23 more rows
```

```r
library(dplyr)

brfss %>%
  filter(!is.na(SDLONELY)) %>%
  distinct(`_STATE`) %>%
  arrange(`_STATE`)
```

```
## # A tibble: 33 x 1
##    `_STATE`
##       <dbl>
##  1        1
##  2        2
##  3        4
##  4        5
##  5        6
##  6       10
##  7       11
##  8       13
##  9       15
## 10       16
## # i 23 more rows
```

```r
library(dplyr)
brfss_subset <- brfss %>%
  filter(!is.na(LSATISFY))%>%
  select(
    MENTHLTH,
    GENHLTH,
    EXERANY2,
    ADDEPEV3,
    MARITAL,
    EDUCA,
    DECIDE,
    DIFFALON,
    LSATISFY,
```

```
    EMTSUPRT ,
    EMPLOY1,
    DRNKANY6,
    SDLONELY,
    HOWSAFE1,
    `_METSTAT`,
    `_RFHLTH`,
    `_PHYS14D`,
    `_TOTINDA`,
    `_RACE`,
    `_SEX`,
    `_AGEG5YR`,
    `_BMI5CAT`,
    `_INCOMG1`,
    `_SMOKER3`,
    `_STATE`
  )
```

```
brfss_subset
```

```
## # A tibble: 203,764 x 25
##    MENTHLTH GENHLTH EXERANY2 ADDEPEV3 MARITAL EDUCA DECIDE DIFFALON LSATISFY
##       <dbl>   <dbl>    <dbl>    <dbl>   <dbl> <dbl>  <dbl>    <dbl>    <dbl>
##  1       88       3        1        2       3     4      2        2        1
##  2       88       1        1        2       1     6      2        2        1
##  3       88       2        1        2       6     5      2        2        1
##  4       88       1        1        2       1     6      2        2        1
##  5       88       3        2        2       5     5      2        1        2
##  6       88       3        1        2       1     4      2        2        1
##  7       88       4        1        2       1     5      2        2        1
##  8       88       5        2        2       1     5      1        1        3
##  9       88       3        1        2       1     4      2        2        1
## 10       88       4        1        2       3     5      2        2        1
## # i 203,754 more rows
## # i 16 more variables: EMTSUPRT <dbl>, EMPLOY1 <dbl>, DRNKANY6 <dbl>,
## #   SDLONELY <dbl>, HOWSAFE1 <dbl>, `_METSTAT` <dbl>, `_RFHLTH` <dbl>,
## #   `_PHYS14D` <dbl>, `_TOTINDA` <dbl>, `_RACE` <dbl>, `_SEX` <dbl>,
## #   `_AGEG5YR` <dbl>, `_BMI5CAT` <dbl>, `_INCOMG1` <dbl>, `_SMOKER3` <dbl>,
## #   `_STATE` <dbl>
```

```
typeof(brfss_subset$`_STATE`)
```

```
## [1] "double"
```

```
brfss_subset <- brfss_subset %>%
  mutate(state_name = recode(as.character(`_STATE`),
    "1"="Alabama", "2"="Alaska", "4"="Arizona", "5"="Arkansas", "6"="California",
    "8"="Colorado", "9"="Connecticut", "10"="Delaware", "11"="District of Columbia",
    "12"="Florida", "13"="Georgia", "15"="Hawaii", "16"="Idaho", "17"="Illinois",
    "18"="Indiana", "19"="Iowa", "20"="Kansas", "21"="Kentucky", "22"="Louisiana",
    "23"="Maine", "24"="Maryland", "25"="Massachusetts", "26"="Michigan",
    "27"="Minnesota", "28"="Mississippi", "29"="Missouri", "30"="Montana",
    "31"="Nebraska", "32"="Nevada", "33"="New Hampshire", "34"="New Jersey",
    "35"="New Mexico", "36"="New York", "37"="North Carolina", "38"="North Dakota",
    "39"="Ohio", "40"="Oklahoma", "41"="Oregon", "42"="Pennsylvania",
```

```
    "44"="Rhode Island", "45"="South Carolina", "46"="South Dakota",
    "48"="Texas", "49"="Utah", "50"="Vermont", "51"="Virginia",
    "53"="Washington", "54"="West Virginia", "55"="Wisconsin", "56"="Wyoming",
    "66"="Guam", "72"="Puerto Rico", "78"="Virgin Islands"
  ))
```

```
brfss_subset
```

```
## # A tibble: 203,764 x 26
##    MENTHLTH GENHLTH EXERANY2 ADDEPEV3 MARITAL EDUCA DECIDE DIFFALON LSATISFY
##       <dbl>   <dbl>    <dbl>    <dbl>   <dbl> <dbl>  <dbl>    <dbl>    <dbl>
##  1       88       3        1        2       3     4      2        2        1
##  2       88       1        1        2       1     6      2        2        1
##  3       88       2        1        2       6     5      2        2        1
##  4       88       1        1        2       1     6      2        2        1
##  5       88       3        2        2       5     5      2        1        2
##  6       88       3        1        2       1     4      2        2        1
##  7       88       4        1        2       1     5      2        2        1
##  8       88       5        2        2       1     5      1        1        3
##  9       88       3        1        2       1     4      2        2        1
## 10       88       4        1        2       3     5      2        2        1
## # i 203,754 more rows
## # i 17 more variables: EMTSUPRT <dbl>, EMPLOY1 <dbl>, DRNKANY6 <dbl>,
## #   SDLONELY <dbl>, HOWSAFE1 <dbl>, `_METSTAT` <dbl>, `_RFHLTH` <dbl>,
## #   `_PHYS14D` <dbl>, `_TOTINDA` <dbl>, `_RACE` <dbl>, `_SEX` <dbl>,
## #   `_AGEG5YR` <dbl>, `_BMI5CAT` <dbl>, `_INCOMG1` <dbl>, `_SMOKER3` <dbl>,
## #   `_STATE` <dbl>, state_name <chr>
```

```
brfss_subset$MENTHLTH[brfss_subset$MENTHLTH %in% c(77,99)] <- NA

brfss_subset$MENTHLTH[brfss_subset$MENTHLTH == 88] <- 0

brfss_subset$`_RFHLTH`[brfss_subset$`_RFHLTH` == 9] <- NA
brfss_subset$`_PHYS14D`[brfss_subset$`_PHYS14D` == 9] <- NA
brfss_subset$`_TOTINDA`[brfss_subset$`_TOTINDA` == 9] <- NA
brfss_subset$MARITAL[brfss_subset$MARITAL == 9] <- NA
brfss_subset$EDUCA[brfss_subset$EDUCA == 9] <- NA
```

```
binary_vars <- c("GENHLTH","EXERANY2","ADDEPEV3","DECIDE","DIFFALON","LSATISFY","EMTSUPRT","SDLONELY","

brfss_subset[binary_vars] <- lapply(brfss_subset[binary_vars], function(x){
  x[x %in% c(7,9)] <- NA
  x
})
```

```
brfss_subset$`_INCOMG1`[brfss_subset$`_INCOMG1` == 9] <- NA
brfss_subset$`_SMOKER3`[brfss_subset$`_SMOKER3` == 9] <- NA
brfss_subset$EMPLOY1[brfss_subset$EMPLOY1 == 9] <- NA
```

```
round(colMeans(is.na(brfss_subset))*100,2)
```

```
##   MENTHLTH    GENHLTH   EXERANY2   ADDEPEV3    MARITAL      EDUCA     DECIDE
##       1.69       0.26       0.24       0.54       0.73       0.38       0.65
##   DIFFALON   LSATISFY   EMTSUPRT    EMPLOY1   DRNKANY6   SDLONELY   HOWSAFE1
##       0.33       0.89       1.54       0.86       1.05       0.80       1.85
##   _METSTAT    _RFHLTH   _PHYS14D   _TOTINDA      _RACE       _SEX   _AGEG5YR
```

```
##       2.64       0.26       2.29       0.24       0.00       0.00       0.00
##   _BMI5CAT   _INCOMG1   _SMOKER3     _STATE state_name
##       5.98      16.89       0.68       0.00       0.00
```

```r
library(dplyr)

brfss_names <- brfss_subset%>%
  mutate(

    # General Health
    GENHLTH = factor(GENHLTH,
                     levels = 1:5,
                     labels = c("Excellent", "Very good", "Good", "Fair", "Poor")),

    # Exercise
    EXERANY2 = factor(EXERANY2,
                      levels = 1:2,
                      labels = c("Yes","No")),

    # Ever told had depressive disorder
    ADDEPEV3 = factor(ADDEPEV3,
                      levels = 1:2,
                      labels = c("Yes","No")),

    # Marital Status
    MARITAL = factor(MARITAL,
                     levels = 1:6,
                     labels = c("Married","Divorced","Widowed","Separated","Never married"," Partnered"

    # Education
    EDUCA = factor(EDUCA,
                   levels = 1:6,
                   labels = c("Never attended","Elementary"," High school","High school graduate","Coll


    # Decision-making Difficulty
    DECIDE = factor(DECIDE,
                    levels = 1:2,
                    labels = c("Yes","No")),

    # Difficulty Doing Errands Alone
    DIFFALON = factor(DIFFALON,
                      levels = 1:2,
                      labels = c("Yes","No")),


    # Life Satisfaction
    LSATISFY = factor(LSATISFY,
                      levels = 1:4,
                      labels = c("Very satisfied","Satisfied","Dissatisfied","Very dissatisfied")),

    # Emotional Support
    EMTSUPRT = factor(EMTSUPRT,
                      levels = 1:5,
```

```
                    labels = c("Always","Usually","Sometimes","Rarely","Never")),
    #Drinking Alcohol
    DRNKANY6 = factor(DRNKANY6,
                    levels = 1:2,
                    labels = c("Yes","No")),


    # Loneliness
    SDLONELY = factor(SDLONELY,
                    levels = 1:5,
                    labels = c("Always","Usually","Sometimes","Rarely","Never")),


    # Neighborhood Safety
    HOWSAFE1 = factor(HOWSAFE1,
                    levels = 1:4,
                    labels = c(" Extremely safe","Safe"," Unsafe","Extremely Unsafe")),


    # Metropolitan Status
    `_METSTAT` = factor(`_METSTAT`,
                     levels = 1:2,
                     labels = c("Metropolitan","Non-Metropolitan")),

    # Race
    `_RACE` = factor(`_RACE`,
                     levels = 1:8,
                     labels = c("White","Black","American Indian/Alaskan Native","Asian","Native Hawa
Islander ","Other","Multiracial"," Hispanic")),

    # Self-rated Health
    `_RFHLTH` = factor(`_RFHLTH`,
                    levels = 1:2,
                    labels = c("Good or better","Fair or poor")),

    # Physical Health Status
    `_PHYS14D` = factor(`_PHYS14D`,
                     levels = 1:3,
                     labels = c("0","1-13","14+")),


    # Physical Activity Indicator
    `_TOTINDA` = factor(`_TOTINDA`,
                     levels = 1:2,
                     labels = c("Yes","No")),


    # Sex
    `_SEX` = factor(`_SEX`,
                    levels = 1:2,
                    labels = c("Male","Female")),

    # Age Group
```

```
    `_AGEG5YR` = factor(`_AGEG5YR`,
                        levels = 1:13,
                        labels = c("18-24","25-29","30-34","35-39","40-44","45-49",
                                   "50-54","55-59","60-64","65-69","70-74","75-79","80+")),

    # BMI Category
    `_BMI5CAT` = factor(`_BMI5CAT`,
                        levels = 1:4,
                        labels = c("Underweight","Normal","Overweight","Obese")),

    # Income
    `_INCOMG1` = factor(`_INCOMG1`,
                        levels = 1:7,
                        labels = c("<$15k","$15-25k","$25-35k","$35-50k","$50k-$100k","$100k-$200k",">$1

    # Smoking Status
    `_SMOKER3` = factor(`_SMOKER3`,
                        levels = 1:4,
                        labels = c("Everyday smoker","Someday smoker","Former smoker","Never smoked")),


EMPLOY1 = factor(EMPLOY1,
                        levels = 1:8,
                        labels = c("Employed for Wages","Self Employed","Out of work for 1 year or more
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.1.6
## v ggplot2   4.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.2.1     v tidyr     1.3.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
brfss_final <- brfss_names%>%
  drop_na(
    MENTHLTH
  )%>% select(-`_STATE`)
```

```
brfss_final
```

```
## # A tibble: 200,326 x 25
##    MENTHLTH GENHLTH   EXERANY2 ADDEPEV3 MARITAL   EDUCA DECIDE DIFFALON LSATISFY
##       <dbl> <fct>     <fct>    <fct>    <fct>     <fct> <fct>  <fct>    <fct>
##  1        0 Good      Yes      No       "Widowed" High~ No     No       Very sa~
##  2        0 Excellent Yes      No       "Married" Coll~ No     No       Very sa~
##  3        0 Very good Yes      No       " Partne~ Coll~ No     No       Very sa~
##  4        0 Excellent Yes      No       "Married" Coll~ No     No       Very sa~
##  5        0 Good      No       No       "Never m~ Coll~ No     Yes      Satisfi~
##  6        0 Good      Yes      No       "Married" High~ No     No       Very sa~
##  7        0 Fair      Yes      No       "Married" Coll~ No     No       Very sa~
##  8        0 Poor      No       No       "Married" Coll~ Yes    Yes      Dissati~
```

```
##  9        0 Good      Yes      No       "Married" High~ No     No       Very sa~
## 10        0 Fair      Yes      No       "Widowed" Coll~ No     No       Very sa~
## # i 200,316 more rows
## # i 16 more variables: EMTSUPRT <fct>, EMPLOY1 <fct>, DRNKANY6 <fct>,
## #   SDLONELY <fct>, HOWSAFE1 <fct>, `_METSTAT` <fct>, `_RFHLTH` <fct>,
## #   `_PHYS14D` <fct>, `_TOTINDA` <fct>, `_RACE` <fct>, `_SEX` <fct>,
## #   `_AGEG5YR` <fct>, `_BMI5CAT` <fct>, `_INCOMG1` <fct>, `_SMOKER3` <fct>,
## #   state_name <chr>
```

```
round(colMeans(is.na(brfss_final))*100,2)
```

```
##   MENTHLTH    GENHLTH   EXERANY2   ADDEPEV3    MARITAL      EDUCA     DECIDE
##       0.00       0.22       0.20       0.47       0.69       0.35       0.58
##   DIFFALON   LSATISFY   EMTSUPRT    EMPLOY1   DRNKANY6   SDLONELY   HOWSAFE1
##       0.29       0.79       1.43       0.81       0.98       0.72       1.78
##   _METSTAT    _RFHLTH   _PHYS14D   _TOTINDA      _RACE       _SEX   _AGEG5YR
##       2.66       0.22       1.83       0.20       1.56       0.00       1.44
##   _BMI5CAT   _INCOMG1   _SMOKER3 state_name
##       5.84      16.54       0.63       0.00
```

```
write.csv(brfss_final,"brfss_names.csv",row.names=FALSE)
```

```
brfss1 <- read.csv("brfss_names.csv")
```

```
summary(brfss1)
```

```
##     MENTHLTH       GENHLTH            EXERANY2           ADDEPEV3
##  Min.   : 0.00   Length:200326      Length:200326      Length:200326
##  1st Qu.: 0.00   Class :character   Class :character   Class :character
##  Median : 0.00   Mode  :character   Mode  :character   Mode  :character
##  Mean   : 4.26
##  3rd Qu.: 4.00
##  Max.   :30.00
##    MARITAL             EDUCA              DECIDE            DIFFALON
##  Length:200326      Length:200326      Length:200326      Length:200326
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    LSATISFY           EMTSUPRT           EMPLOY1            DRNKANY6
##  Length:200326      Length:200326      Length:200326      Length:200326
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    SDLONELY           HOWSAFE1          X_METSTAT           X_RFHLTH
##  Length:200326      Length:200326      Length:200326      Length:200326
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##   X_PHYS14D          X_TOTINDA            X_RACE             X_SEX
##  Length:200326      Length:200326      Length:200326      Length:200326
```

```
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##   X_AGEG5YR          X_BMI5CAT          X_INCOMG1          X_SMOKER3
##  Length:200326      Length:200326      Length:200326      Length:200326
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##   state_name
##  Length:200326
##  Class :character
##  Mode  :character
##
##
##
```